Multi-level factors associated with low dose computed tomography lung cancer screening in the United States

Tung-Sung Tseng, Chien-Ching Li, Hui-Yi Lin, Kelsey N. Witmeier, Chaoyi Zeng, Yu-Wen Chiu, Michael D. Celestin, Edward J. Trapido

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Abstract: Low-dose computed tomography (LDCT) screening is recommended for high-risk smokers to decrease lung cancer-related mortality and increase prognosis. In the U.S., the uptake of LDCT screening among eligible smokers is suboptimal. The impacts of social-environmental and individual factors on LDCT screening uptake using nationally representative dataset recommendations are understudied. The current study investigated multi-level factors associated with LDCT screening using national data.

Methods: The 2017-2021 Behavioral Risk Factor Surveillance System (BRFSS) data, social determinants of health (SDOH) and other state-level variables (Medicaid expansion status and the number of screening facilities using American Lung Association (ALA) 's State of Lung Cancer) were applied. Our study outcome variance was LDCT screening among study participants who met the U.S. Preventive Services Task Force guidelines for lung cancer screening. The final study sample consisted of 15640 respondents from 29 states. All analyses were weighted to account for the complex sampling design applied in BRFSS.

Results: The overall utilization rate of LDCT screening is only 18.4%. The LDCT screening rate varied by state (6.2 -31.1%). LDCT screen rates were not significantly associated with the number of lung cancer screening facilities ($r=0.02$, $p=0.909$) but were positively associated with the number of lung cancer screening facilities per 10,000 smokers ($r=0.67$, $p<0.001$). Among the respondents, individuals who were employed, never married, reported good health status, did not have primary care physicians, economic concerns like low income, did not have routine checkups, and did not have certain chronic conditions (i.e. cancer, asthma, COPD) had a lower utilization rate of LDCT screening compared with their counterpart.

Conclusion: The use of LDCT screening among eligible smokers remains low. Enhancing access to care for high-risk individuals, promoting services to diverse racial and socioeconomic groups, and expanding Medicaid coverage to incorporate annual LDCT screening can be used to guide future lung cancer screening programs.

Keywords: Cancer ■ Tobacco ■ Screening ■ Behaviors ■ Surveillance

Author affiliations: Tung-Sung Tseng School of Public Health, Louisiana State University Health Sciences Center, USA; Chien-Ching Li Department of Health Systems Management, Rush University, USA; Hui-Yi Lin School of Public Health, Louisiana State University Health Sciences Center, USA; Kelsey N. Witmeier School of Public Health, Louisiana State University Health Sciences Center, USA; Chaoyi Zeng School of Public Health, Louisiana State University Health Sciences Center, USA; Yu-Wen Chiu School of Public Health, Louisiana State University Health Sciences Center, USA; Michael D. Celestin School of Public Health, Louisiana State University Health Sciences Center, USA; Edward J. Trapido School of Public Health, Louisiana State University Health Sciences Center, USA

Corresponding author: Tung-Sung Tseng, email: ttseng@lsuhsc.edu

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INTRODUCTION

Lung cancer (LC) is the third most prevalent and second most deadly cancer in the U.S.[1] In 2020, around 197,500 people were diagnosed, and 136,00 people died from LC.[1] The five-year survival rate for LC is roughly 26%.[1] Early detection of lung cancer through screening significantly improves survival rates.[1] Individuals diagnosed at an early, localized stage experience considerably higher survival compared to those diagnosed at later, more advanced stages.[1] However, those with an early diagnosis (localized cancer) have a longer survival than those diagnosed in later stages.[1]

Early detection of LC through low-dose computed tomography (LDCT) screening significantly reduces lung cancer mortality, with studies demonstrating a a 21% mortality reduction in high-risk individuals.[2] This mortality reduction may be a result of the tool's capacity to identify cancer during Stage 1 instead of at a later stage when cancer has spread.[3]

The current U.S. Preventive Services Task Force screening guidelines recommend annual LDCT screening for individuals aged 50-80 who have at least a 20-pack-year smoking history and current clock or have quit within the past 15 years.[4] These guidelines reflect updated recommendations based on recent research and improved modeling to better predict those who may benefit from LDCT screening.[4]

Currently, LDCT screening has a low uptake among eligible individuals. LDCT screening rates of eligible individuals have been found to range from 5% to 19%. [5,6] Between 2010 and 2015, LDCT screening among eligible participants only increased by 0.6% (3.3% to 3.9%)[7]; the screening rate rose from 1 in 1,000 eligible individuals to 6 in 1,000 from 2016 to 2017.[8]

Multiple-ecological factors ranging from individual characteristics, and policies are influential in LDCT screening uptake.[9-12] At an individual level, LDCT screening utilization has been associated with an individual's demographic factors including race, employment status, income, and physician attainment status.[5,9,13] Specifically,

individuals who were retired or unable to work, White, had an established primary care provider (PCP), made at least $10,000 annually, and did not have a high school diploma had a higher likelihood of obtaining annual screening.[5,9,13] Positive intrapersonal risk factors for screening include attitudes and beliefs toward the screening tool, motivation to understand one's health, and knowledge about the screening tool.[11,14] Screening rates also varied by state-twenty-two states had rates below the national average, with most in the Southern and Western US.[6] Geographical differences in access to the screening, insurance coverage, barriers to scheduling and getting to these appointments, and financial burdens have been linked to annual LDCT screening.[11,12,14,15] Policy factors such as variations in state Medicaid coverage of LDCT screenings, and Medicare's enrollment criteria further limits access to and the use of LDCT screenings.[11,16] According to the 2023 report from American Lung Association, one state's Medicaid program does not cover LDCT screening for eligible individuals, and 23 states are not using updated guidelines.[17]

LOne study found that 28.1% of individuals aged 55 to 79 did not have an LDCT center within 30 minutes of them.[18] Those in rural areas have poorer access to screening centers than their counterparts.[18] This can result in differential identification rates of LC among those in varied geographic locations.[18] Challenges in obtaining and receiving the screening also pose an issue. People must take off work, have transportation to the screening, and seek childcare to get an LDCT.[14] This can result in a loss of income and may be a significant inconvenience.[14] Perceived financial burden may also be a deterrent if people feel they may have to pay out of pocket.[14]

Current research evaluates cohorts within singular or multiple states which may not be representative of those eligible from a national level. Most research also centers on individual-level factors instead of evaluating multi-level factors. There is a need for a national evaluation of barriers and factors associated with LDCT screening uptake to better guide programs to either reinforce or improve existing recruitment techniques.The current study investigated multi-level factors associated with utilizing LDCT screening using national data.

METHOD

Study population

This study used data from the 2017-2021 Behavioral Risk Factor Surveillance System (BRFSS, n=15,640), which had LDCT screening data from 29 states in the U.S. The BRFSS is a U.S. population-based and cross-sectional survey, collected in 29 states. The inclusion criteria for this study are are (1) aged 55 to 79, (2) with a 30-pack-year smoking history who currently smokes or has quit smoking within the past 15 years, and (3) having health insurance. The pack-year smoking history was calculated by multiplying the number of packs of cigarettes smoked per day by the number of years the person has smoked. A standard pack of cigarettes typically contains 20 cigarettes. The first two criteria were primarily set up based on the 2013 USPSTF guidelines who had health insurance. Based on the 2013 USPSTF guideline, persons aged 55 to 80 with a 30-pack-year smoking history who currently smoke or have quit smoking within the past 15 years are advised to have an annual lung cancer screening with the LDCT.[4] Although the LDCT eligibility criteria were revised in 2021, we applied the 2013 guideline because this study used the BRFSS data on and before 2021. Among those who were suggested for LDCT lung cancer screening based on the 2013 USPSTF guidelines, only a small number of individuals (6%) did not have health insurance. Therefore, this study only included those with health insurance. Moreover, the original age variable from the BRFSS data was based on 5-year intervals (such as 55-59, 60-64, .75-79, and 80+), so participants aged 55-79 were included in this study. This study is based on a secondary data analysis using open-access BRFSS data, so informed consent is not applicable.

Statistical analyses

This study has two parts: individual-level and state-level analyses. For the individual-level analyses, the 2017-2021 BRFSS data were analyzed. The outcome was LDCT screening (yes/no). The predictors of interest included demographic factors (such as age, sex, and race), health behaviors and health care factors (such as cigarette smoking and alcohol consumption, personal physician and routine checkups), and health-related factors (health status and medical conditions such as any type of cancer other than lung cancer, asthma and chronic obstructive pulmonary disease (COPD). The factors were stratified by LDCT screening using descriptive statistics. To evaluate associations between each factor and LDCT screening, we applied the Rao-Scott chi-square test. The association between each factor and LDCT use status with and without adjusting other factors was tested using univariate and multivariable weighted logistic models. The factors in the final multivariable logistic model include basic demographic factors (age, sex, and race) and factors with a p-value less than 0.1 in the univariate models. Two factors (personal physician and routine checkups within the past year) were

highly correlated (p<0.001) so only the personal physician was considered in the final multivariable logistic model.

For smoking status, never smokers were defined as those who smoked less than 100 cigarettes in their lifetime, former smokers were defined as those who smoked at least 100 cigarettes in their lifetime but did not smoke currently, and current smokers were defined as those who smoked at least 100 cigarettes in their lifetime and now smoked every day or some days. The never smokers were excluded from this study based on the LDCT eligibility criteria. The adjusted odds ratios (ORs) and the corresponding 95% confidence intervals (CIs) of these factors were calculated. The statistical significance level was set at 0.05. All analyses were weighted to account for the complex sampling design applied in BRFSS using SAS 9.4 (SAS Institute, Inc., Cary, North Carolina). The weighting analytical procedures followed the suggested instructions.

For the state-level data analyses, the LDCT screening and smoking rates for 2021 were obtained from the American Lung Association (ALA) 's State of Lung Cancer report.[19] The number of smokers for each state was estimated using the state-level smoking rate multiplied by the number of adults in each state. The number of adults aged 18 and older by state was obtained from the U.S. 2020 Census data.[20] The number of lung cancer screening facilities by state in 2021 was obtained based on the Lung Cancer Screening Registry (LCSR) state reports from the National Radiology Data Registry.[21] The coverage status (yes/no) of Medicaid fee-for-service programs (Medicaid Expansion) that covered lung cancer screening by state was also included. The associations between the state-level LDCT screening rates and lung cancer screening facilities were evaluated using the Pearson correlation and linear regression. The association between the state-level LDCT screening rates and Medicaid Expansion coverage status was tested using the t-test.

RESULTS

Our study analyzed the individual-level BRFSS data collected in 29 states from 2017 to 2021. A total of 15,640 residents from 29 states were included. The weighted lung cancer screening rate was 18.4% among individuals eligible for LDCT screening based on the 2013 USPSTF LDCT screening guidelines. Our results showed that several demographic factors, health status, health behavior factors, and health-related factors were associated with LDCT screening use. In terms of demographic factors (Table 1), individuals who were employed (12.7%) or never married (13.5%) had lower LDCT screening rates compared with their counterparts (p<0.05). Among health behaviors and health care factors (Table 2), no personal physician (9.1%), not seeing a doctor because of cost problems (13.7%), not having routine checkups (1.2%), and no flu shot in last year (15.9%) were associated with lower LDCT screening (p<0.05). Regarding health-related factors (Table 3), individuals reporting "very good health" (14%), "no unhealthy days in the past 30 days" (14.9%), or not having cancer (16.2%), asthma (17.1%), and COPD (12.7%) had lower LDCT screening use (p<0.05). Table 4 shows the results from the multivariable logistic model including demographic factors (age, sex, race) and factors with a p-value less than 0.1 in the univariate models. Individuals aged 75-79 years (OR=0.46, 95% confidence interval (CI)=0.24-0.86, p=0.015) or reported not seeing a doctor because of cost problems (OR=0.66, 95% CI=0.45-0.97, p=0.035) were less likely to use LDCT screening compared to their reference group. Instead, those who had a personal physician (OR=2.03, 95% CI=1.19-3.46, p=0.009), received flu vaccine (OR=1.28, 95% CI=1-1.63, p=0.047), had any cancer (OR=1.62, 95% CI=1.29-2.05, p<0.001) or had COPD (OR=2.29, 95% CI=1.75-2.99, p<0.001) were more likely to use LDCT screening.

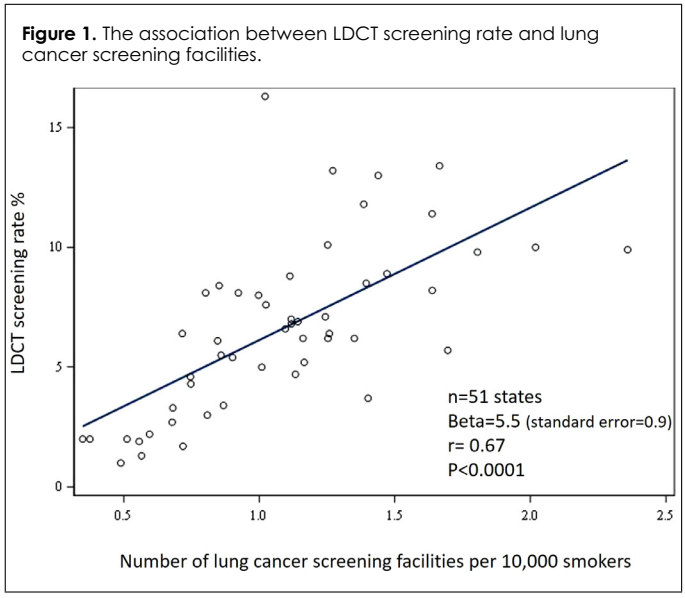


Figure 1. The association between LDCT screening rate and lung cancer screening facilities.

For the state-level data analyses (Figure 1), LDCT screen rates were not significantly associated with the number of lung cancer screening facilities (r=0.02, p=0.909) but were positively associated with the number of lung cancer screening facilities per 10,000 smokers (r=0.67, p<0.001).

DISCUSSION

Despite the recommendations from the USPSTF, the screening rate for LDCT remains low in the U.S.[22, 23] The

Table 1. Demographic factors associated with LDCT[1] screening (BRFSS 2017-2021).

Factors	LDCT Screening[1] N (wt%)	No LDCT Screening[1] N (wt%)	p-value[2]
Total	2604 (18.4)	13036 (81.6)	
Age			
55-59	390 (15.5)	2702 (84.5)	0.256
60-64	635 (17.7)	3434 (82.3)	
65-69	724 (21.0)	3142 (79.0)	
70-74	587 (21.0)	2396 (79.0)	
75-79	268 (16.9)	1362 (83.1)	
Sex			
Female	1230 (20.0)	6046 (80.0)	0.125
Male	1372 (17.0)	6987 (83.0)	
Race			
White	2313 (18.7)	11560 (81.3)	0.777
Black	74 (14.7)	427 (85.3)	
Other	170 (18.2)	844 (81.8)	
Education			
Did not graduate high school	264 (18.6)	1283 (81.4)	0.995
Graduated high school	944 (18.1)	4992 (81.9)	
Attended college/technical school	840 (18.6)	4228 (81.4)	
Graduated from college	551 (18.0)	2513 (82.0)	
Employment status			
Employed	557 (12.7)	3859 (87.3)	**<0.001**
Unemployed	697 (20.3)	3039 (79.7)	
Retired	1344 (21.3)	6103 (78.7)	
Marital status			
Married or living with a partner	1174 (17.1)	6216 (82.9)	**0.028**
Never married	184 (13.5)	920 (86.5)	
Divorce/widow/separate	1235 (20.8)	5849 (79.2)	
Income category			
<$20,000	607 (19.1)	2663 (80.9)	0.221
$20,000 to <$50,000	923 (20.7)	4633 (79.3)	
>$50,000	721 (16.4)	3783 (83.6)	

[1] *LDCT: low-dose computed tomography.*

low rate of LDCT screening is concerning as the screening is a covered service under Medicare, most state Medicaid fee-for-service programs, and most private insurance plans because of the USPSTF recommendations and provisions of the Patient Protection and Affordable Care Act.[17]

Current literature estimates screening rates among eligible individuals to vary from 5.8% to 19%.[5, 19,35] Findings differ regarding the sample population, such as high-risk group, eligibility, or general population and the data collection methods used in the respective study's analysis.[23]

Table 2. Health behaviors and health care factors associated with LDCT Screening (BRFSS 2017-2021).

Factors	LDCT Screening N (wt%)	No LDCT Screening N (wt%)	p-value
Current smoker			
No	1398 (19.6)	6739 (80.4)	0.220
Yes	1206 (17.2)	6297 (82.8)	
Smoking year for ever smokers			
30-39.9	418 (16.1)	3131 (83.9)	0.056
40- 44.9	596 (16.2)	3304 (83.8)	
45- 49.9	676 (18.7)	2981 (81.3)	
≥50	914 (22.5)	3620 (77.5)	
Pack-year			
30—39.9	509 (15.0)	3490 (85.0)	0.092
40- 49.9	814 (18.0)	4234 (82.0)	
50- 59.9	498 (20.0)	2091 (80.0)	
≥60	783 (21.6)	3221 (78.4)	
Alcohol Consumption			
Non-drinker	1520 (19.2)	7182 (80.8)	0.6209
Non-heavy drinker	806 (17.3)	4433 (82.7)	
Heavy drinker	231 (18.5)	1148 (81.5)	
Has personal physician			
No	119 (9.1)	1253 (90.9)	<0.001
Yes	2480 (19.4)	11743 (80.6)	
Could not see a doctor because of cost problem			
No	2400 (18.9)	11796 (81.1)	**0.022**
Yes	197 (13.7)	1208 (86.3)	
Routine checkup			
Never	3 (1.2)	71 (98.8)	**<.0001**
Within past year	2440 (20.3)	10984 (79.7)	
Within the past 2 years	96 (7.9)	925 (92.1)	
Within the past 5 years	30 (6.4)	430 (93.6)	
5 or more years ago	20 (3.0)	513 (97.0)	
Flu vaccine			
No	915 (15.9)	6215 (84.1)	**0.019**
Yes	1675 (20.7)	6770 (79.3)	

[1] *LDCT: low-dose computed tomography.*

In this study, we found that 18.4% of eligible individuals received LDCT screening from 2017-2021 under 2013 USPSTF guidelines. Our findings reported similar results (18.4% vs. 19.2%) with Narayan's findings from 8 states.[5]

Several health system, clinician, and patient level factors contribute to low screening rates. At the system level, lack of policies to disseminate the LDCT screening more broadly, clinical readiness to implement screen-

Table 3. Health-related factors associated with LDCT screening (BRFSS 2017-2021).

Factors	LDCT Screening N (wt%)	No LDCT Screening N (wt%)	p-value
Health status			
Very good	552 (14.0)	3806 (86.0)	**0.001**
Good	791 (16.8)	4561 (83.2)	
Fair	745 (21.4)	3019 (78.6)	
Poor	506 (25.6)	1610 (74.4)	
Number of Days Physical Health Not Good in past 30 days			
0	1052 (14.9)	6707 (85.1)	**<0.001**
1 to 15	664 (19.2)	3171 (80.8)	
>15	805 (24.6)	2833 (75.4)	
Any cancer			
No	1644 (16.2)	9992 (83.8)	**<.0001**
Yes	960 (25.1)	3044 (74.9)	
Asthma			
No	2058 (17.1)	11110 (82.9)	**0.004**
Yes	531 (25.5)	1871 (74.5)	
COPD			
No	1151 (12.7)	8557 (87.3)	**<.0001**
Yes	1434 (28.3)	4362 (71.7)	
Heart attack			
No	2109 (18.2)	10864 (81.8)	0.607
Yes	477 (19.4)	2092 (80.6)	
CHD			
No	2094 (18.2)	10964 (81.8)	0.767
Yes	456 (18.8)	1825 (81.2)	
Stroke			
No	2294 (18.4)	11728 (81.6)	0.993
Yes	298 (18.4)	1262 (81.6)	
Arthritis			
No	1109 (16.8)	6299 (83.2)	0.104
Yes	1482 (20.0)	6647 (80.0)	
Depressive Disorder			
No	1766 (17.7)	9484 (82.3)	0.159
Yes	829 (20.6)	3509 (79.4)	
Kidney Disease			
No	2401 (18.2)	12191 (81.8)	0.531
Yes	184 (20.7)	795 (79.3)	
Diabetes			
No	1978 (17.5)	9971 (82.5)	0.103
Yes	621 (21.2)	3042 (78.8)	

[1] LDCT: low-dose computed tomography.

Table 4. Factors associated with LDCT screening (BRFSS 2017-2021): Logistic regression results.

Factors	Unadjusted OR (95% CI) of LDCT Screening	p-value	Adjusted OR (95% CI) of LDCT Screening	p-value
Age				
55-59	1		1	
60-64	1.18 (0.78, 1.77)	0.435	0.95 (0.65, 1.4)	0.799
65-69	1.45 (0.97, 2.17)	0.070	0.84 (0.54, 1.3)	0.433
70-74	1.46 (0.96, 2.21)	0.076	0.72 (0.43, 1.21)	0.217
75-79	1.11 (0.64, 1.94)	0.704	**0.46 (0.24, 0.86)**	**0.015**
Sex				
Female	1		1	
Male	0.82 (0.63, 1.06)	0.126	0.99 (0.77, 1.28)	0.935
Race				
White	1		1	
Black	0.75 (0.39,1.45)	0.391	0.71 (0.39, 1.29)	0.260
Other	0.97 (0.44, 2.13)	0.933	1.03 (0.49, 2.13)	0.947
Employment status				
Employed	1		1	
Unemployed	**1.75 (1.24, 2.48)**	**0.002**	1.2 (0.8, 1.79)	0.384
Retired	**1.86 (1.34, 2.58)**	**<.0001**	1.46 (0.98, 2.19)	0.065
Marital status				
Married or living with a partner	1		1	
Never married	0.76 (0.53, 1.10)	0.151	0.75 (0.49, 1.14)	0.172
Divorce/widow/separate	1.28 (0.98, 1.67)	0.074	1.21 (0.91, 1.6)	0.195
Smoking year for ever smokers				
30-39.9	1		1	
40- 44.9	1.00 (0.66, 1.52)	0.987	1.05 (0.73, 1.52)	0.798
45- 49.9	1.20 (0.79, 1.81)	0.398	1.01 (0.7, 1.46)	0.959
≥50	1.51 (0.99, 2.30)	0.053	1.34 (0.91, 1.98)	0.136
Health status				
Very good	1		1	
Good	1.25 (0.83, 1.87)	0.283	1 (0.7, 1.41)	0.985
Fair	**1.68 (1.11, 2.54)**	**0.014**	1.27 (0.86, 1.87)	0.227
Poor	**2.12 (1.40, 3.21)**	**<0.001**	1.32 (0.85, 2.04)	0.218
Has personal physician				
No	1		1	
Yes	**2.40 (1.49, 3.87)**	**<.0001**	**2.03 (1.19, 3.46)**	**0.009**

(*continued on next page*)

Table 4 *(continued)*

Factors	Unadjusted OR (95% CI) of LDCT Screening	p-value	Adjusted OR (95% CI) of LDCT Screening	p-value
Could not see a doctor because of cost problem				
No	1		1	
Yes	**0.69 (0.49, 0.95)**	**0.024**	**0.66 (0.45, 0.97)**	**0.035**
Flu vaccine				
No	**1**		1	
Yes	**1.38 (1.05, 1.81)**	**0.019**	**1.28 (1, 1.63)**	**0.047**
Any cancer				
No	**1**		1	
Yes	**1.74 (1.35, 2.23)**	**<.0001**	**1.62 (1.29, 2.05)**	**<.0001**
Asthma				
No	**1**		1	
Yes	**1.66 (1.17, 2.36)**	**0.005**	1.17 (0.79, 1.73)	0.426
COPD				
No	**1**		1	
Yes	**2.73 (2.11, 3.54)**	**<.0001**	**2.29 (1.75, 2.99)**	**<.0001**

[1] *LDCT: low-dose computed tomography; Adjusted for states.*

ing, and incentives for systems and clinicians hinder uptake of LDCT screening.[24] At the clinician level, LDCT screening knowledge and beliefs, and perceptions of the strength of evidence for LDCT screeninghinder uptake. [24]At the patient level, major barriers include having limited or no access to preventive care (such as regular check-ups and having a PCP) and financial barriers which hinder individuals from seeking necessary care.[25] Our findings and previous literature support the association between preventative care and LDCT screening. In our current study, low LDCT screening rates were significantly associated with poor healthcare behaviors or health-related factors, including not having a PCP, not receiving a routine check-up, or not getting a flu vaccine. Narayan and their team reported similar findings; having an established PCP was statistically significantly associated with increased LDCT screening rates.[5] A PCP may provide information on LDCT screening information to eligible patients and motivational interviewing to reluctant individuals.[10] Physicians are recommended to use a shared decision-making model to facilitate patient empowerment, which, in turn, may increase the LDCT screening rates.[26,27] Increasing access to and connections with preventative services among those at risk for lung cancer may improve LDCT screening.

Financial barriers inhibit individuals from LDCT screening. Our study found that not going to a doctor because of a cost issue was significantly associated with low LDCT uptake. Other research has outlined financial and situational concerns as barriers to LDCT screening.[11,12] These financial and situational concerns may include costs to get to appointments, being unable to miss work because they will lose wages, etc. Decreasing financial barriers may increase LDCT screening rates. Individuals who have never been married or employed frequently encounter social and economic disadvantages, which may negatively influence their healthcare access and health literacy. Wong et.al.found that individuals eligible for LDCT screening who had never been married, no perceived risk or family history of lung cancer, no personal history of emphysema, and current smokers were less likely to receive LDCT screening, compared to their counterparts.[27] Additionally, employed individuals, particularly those in low-wage jobs or without access to paid time off, may struggle to find

the time to schedule and attend medical appointments. Finally, knowledge of LDCT is significantly associated with uptake.[25]

Individual characteristics were associated with LDCT screening among eligible respondents. Specifically, our study found that individuals who did not have health conditions like COPD, were never married, and were employed were less likely to receive LDCT screening. These findings are consistent with other research. Narayan and their research team found that individuals who were single and employed were less likely to receive LDCT screening.[5] Findings from Zgodic et al. and Yong et al. support that individuals with COPD had increased odds of receiving LDCT screening when compared to those without COPD.[28,29] As expected, those with comorbidities were more likely to receive screening as these individuals are likely to be connected with a PCP and seek necessary treatment for their other conditions. Our study found that income and race were not significantly associated with screening uptake. Research conducted by Yong and their research team did not find a significant association between race and LDCT screening as well. [29] Alternatively, previous research pointed to racial and socioeconomic disparities in LDCT uptake; people of color and those who receive less than 10,000 dollars a year had poorer screening rates than their counterparts.[9] Our findings may differ from other study results because of the definition of screening eligibility used and the population covered, the availability of LDCT screening facilities, the level of awareness of LDCT by PCPs, and out of pocket costs even when services are covered by insurance.

Furthermore, we observed considerable variation in LDCT screening rates among the 29 states evaluated, ranging from 6.2% in Nevada to 31.1% in Texas. In addition to individual-level factors, state-level factors such as availability and accessibility of certified LDCT facilities and Medicaid expansion policy may be contributing factors to low LDCT screening rate. Our findings indicated that LDCT screening rates were not significantly associated with the number of lung cancer screening facilities ($r=0.02$, $p=0.909$). However, the number of lung cancer screening facilities per 10,000 smokers was significantly associated with LDCT screening rates. Additionally, there was no significant association between LDCT screening rates and state-level Medicaid expansion coverage status ($p=0.494$). This is consistent with findings from the study conducted by Fedewa et al.[6] Financial barriers, such as cost-sharing, could impede high-risk Medicaid enrollees from accessing LDCT screening.

The study adds to the literature on LDCT screening in the U.S, and the influence of varying factors on screening uptake. Specifically, this study evaluated LDCT screening rates from a national and state level. This study included data from four years (2017-2021) and a representative sample of the U.S. population. The data included may increase the finding's generalizability and representativeness of the population of interest. We also included numerous factors in our analysis to better understand the influence of different variables/factors. We could better understand the individual influence of factors on LDCT screening before and after controlling for the other covariates.

Despite the strengths of the study, there were limitations. The current study analyzed the 2017-2021 BRFSS using the 2013 USPSTF standard. Under this recommendation, individuals eligible for LDCT screening included individuals aged 55-80 with a 30-pack-year smoking history who are current smokers or quit in the past 15 years.[4] In 2021, the USPSTF recommendation was updated to reach more individuals at risk for lung cancer. This research-supported guideline decreased the pack-year requirement to 20 and increased the eligible age gap from 55-80 to 50-80 years.[4] This updated recommendation is expected to decrease the prevalence of LDCT screening among eligible individuals because of eligible population increased. Racial disparities are also expected to widen the expansion of the LDCT screening guidelines. In future research, the BRFSS from 2021 until the present should be analyzed using the new standard to evaluate overarching uptake and disparities. We also are unable to assess other individual-level (e.g., attitude and knowledge) and state-level factors (e.g., number of smoking cessation programs and healthcare providers) that may be the most influential. This may pose difficulty in identifying the best target in future programs. However, other research will inform us on the temporality and causative factors.

CONCLUSION

LDCT screening was associated with inadequate access to routine health care, a personal physician, and financial burdens accompanied with seeking healthcare. Enhancing access to care for high-risk individuals, promoting services to diverse racial and socioeconomic groups, and expanding Medicaid coverage to incorporate annual low-dose C.T. scans for all high-risk Medicaid enrollees without cost-sharing could play a pivotal role in elevating LDCT screening uptake in the U.S.

REFERENCES

1. Centers for Disease Control and Prevention. *Lung Cancer Statistics* June 8. U.S. Department of Health and Human Services and U.S. Cancer Statistics; 2023 Retrieved from. https://www.cdc.gov/cancer/lung/statistics/index.htm

2. Bonney A, Malouf R, Marchal C, et al. Impact of low-dose computed tomography (LDCT) screening on lung cancer-related mortality. *Cochrane Database Syst Rev*. 2022(8).
3. Qiao EM, Voora RS, Nalawade V, et al. Evaluating the clinical trends and benefits of low-dose computed tomography in lung cancer patients. *Cancer Med*. 2021;10(20):7289–7297. doi:10.1002/cam4.4229.
4. Krist AH, Davidson KW, Mangione CM, et al.U.S. Preventive Services Task Force Screening for lung cancer: U.S. Preventive Services Task Force recommendation statement. *JAMA*. 2021;325(10):962–970. doi:10.1001/jama.2021.1117.
5. Narayan AK, Gupta Y, Little BP, Shepard JO, Flores EJ. Lung cancer screening eligibility and use with low-dose computed tomography: results from the 2018 Behavioral Risk Factor Surveillance System cross-sectional survey. *Cancer*. 2021;127(5):748–756. doi:10.1002/cncr.33322.
6. Fedewa SA, Kazerooni EA, Studts JL, et al. State variation in low-dose computed tomography scanning for lung cancer screening in the United States. *J Natl Cancer Inst*. 2021;113(8):1044–1052.
7. Jemal A, Fedewa SA. Lung cancer screening with low-dose computed tomography in the United States—2010 to 2015. *JAMA Oncol*. 2017;3(9):1278–1281.
8. Okereke IC, Nishi S, Zhou J, Goodwin JS. Trends in lung cancer screening in the United States, 2016–2017. *J Thorac Dis*. 2019;11(3). doi:10.21037/jtd.2019.01.105.
9. Sosa E, D'Souza G, Akhtar A, et al. Racial and socioeconomic disparities in lung cancer screening in the United States: a systematic review. *Cancer J Clin*. 2021;71(4):299–314.
10. Wiener RS, Gould MK, Arenberg DA, et al. An official American Thoracic Society/American College of Chest Physicians policy statement: implementation of low-dose computed tomography lung cancer screening programs in clinical practice. *Am J Respir Crit Care Med*. 2015;192(7):881–891.
11. Bernstein E, Bade BC, Akgün KM, Rose MG, Cain HC. *Seminars in Oncology*. Barriers and facilitators to lung cancer screening and follow-up. WB Saunders; 2022.
12. Shin D, Fishman MD, Ngo M, Wang J, LeBedis CA. The impact of social determinants of health on lung cancer screening utilization. *J Am Coll Radiol*. 2022;19(1):122–130.
13. Richmond J, Mbah OM, Dard SZ, et al. Evaluating potential racial inequities in low-dose computed tomography screening for lung cancer. *J Natl Med Assoc*. 2020;112(2):209–214.
14. Cavers D, Nelson M, Rostron J, et al. Understanding patient barriers and facilitators to uptake of lung screening using low dose computed tomography: a mixed methods scoping review of the current literature. *Respir Res*. 2022;23(1):1–18.
15. Sakoda LC, Rivera MP, Zhang J, et al. Patterns and factors associated with adherence to lung cancer screening in diverse practice settings. *JAMA Netw*. 2021;4(4).
16. Patel DC, He H, Berry MF, et al. Cancer diagnoses and survival rise as 65-year-olds become Medicare-eligible. *Cancer*. 2021;127(13):2302–2310.
17. American Lung Association. *State of Lung Cancer Screening Coverage Toolkit: Lung Cancer Screening Coverage in State Medicaid Fee-for-Service Programs* October. American Lung Association; 2023 Retrieved from. https://www.lung.org/lung-health-diseases/lung-disease-lookup/lung-cancer/screening-resources/state-lung-cancer-screening
18. Eberth JM, Bozorgi P, Lebrón LM, et al. Peer reviewed: geographic availability of low-dose computed tomography for lung cancer screening in the United States, 2017. *Prev Chronic Dis*. 2018:15.
19. American Lung Association. *State of Lung Cancer 2021 Report*. American Lung Association; 2021 Retrieved from. https://www.lung.org/getmedia/ba972351-ddc5-46b2-8e0d-028002d16c72/solc-2021-print-report-final.pdf
20. U.S. Census Bureau. *National Population by Characteristics: 2020-2023* December 18. U.S. Census Bureau; 2023 Retrieved from:. https://www.census.gov/data/tables/time-series/demo/popest/2020s-national-detail.html
21. American College of Radiology. *Lung Cancer Screening Registry State Level Comparisons*. American College of Radiology; 2023 Retrieved from:. https://www.acr.org/Practice-Management-Quality-Informatics/Registries/Lung-Cancer-Screening-Registry
22. Pham D, Bhandari S, Pinkston C, Oechsli M, Kloecker G. Lung cancer screening registry reveals low-dose CT screening remains heavily underutilized. *Clin lung cancer*. 2020;21(3):e206–e211.
23. Pinsky PF, Miller E. Use and outcomes of low-dose CT scan lung cancer screening in the Medicare population. *Chest*. 2022;162(3):721–729.
24. Sedani AE, Davis OC, Clifton SC, Campbell JE, Chou AF. Facilitators and barriers to implementation of lung cancer screening: a framework-driven systematic review. *J Natl Cancer Inst*. 2022;114(11):1449–1467. doi:10.1093/jnci/djac154.
25. Doubeni CA, Laiyemo AO, Young AC, et al. Primary care, economic barriers to health care, and use of colorectal cancer screening tests among Medicare enrollees over time. *Ann fam med*. 2010;8(4):299–307. doi:10.1370/afm.1112.
26. Nishi SP, Lowenstein LM, Mendoza TR, et al. Shared decision–making for lung cancer screening: how well are we "sharing"? *Chest*. 2021;160(1):330–340.
27. Wong LY, Choudhary S, Kapula N, et al. Barriers to completing low dose computed tomography scan for lung cancer screening. *Clin Lung Cancer*. 2024;25(5):424–430. doi:10.1016/j.cllc.2024.04.014.
28. Zgodic A, Zahnd WE, Miller Jr DP, Studts JL, Eberth JM. Predictors of lung cancer screening utilization in a population-based survey. *J Am Coll Radiol*. 2020;17(12):1591–1601.
29. Yong PC, Sigel K, Rehmani S, Wisnivesky J, Kale MS. Lung cancer screening uptake in the United States. *Chest*. 2020;157(1):236–238.